

Modelling a ferromagnet using the Ising model

By Charlie Cook

Ferromagnets can be modelled using a simple **2D Ising model**. Using this model, and **Monte-Carlo methods** (the **metropolis algorithm**), we determine the **average magnetisation** in the **steady state region** for a set of specific conditions and compare to other models of **ferromagnets**, all showing good agreement with one another.

Introduction

- Magnetism** has long been central to technology, from early compasses to modern data storage.
- Many models successfully described **ferromagnetism**, such as **Mean Field Theory (MFT)** and the **Heisenberg model**, but the **Ising model** stands out for its **simplicity** and **analytical tractability**.
- Despite being simple, it captures key phenomena, such as **phase transitions and critical temperatures**, and is widely used even outside of physics e.g. to model financial markets.
- In this project, the **Ising model** is used to calculate the **average magnetisation** (at steady state) of a lattice when placed in an **external field**

1. 2D Ising Model

- A **periodic lattice** of dimensions N_x by N_y is created.
- Each point is randomly assigned a **spin** value, S , of either +1 (**spin up**) or -1 (**spin down**).
- The **total energy** of the lattice comes from **two sources** [1,2] and is then given by:

$$E = -B \sum_{\alpha} S_{\alpha} - J \sum_{\langle \alpha \beta \rangle} S_{\alpha} S_{\beta}$$

External Field Contribution Spin Interaction Contribution

- where α is the summation over each **lattice point** (i.e. over all i and j), and β is the summation of the **four nearest neighbours**: $(i-1, j)$, $(i+1, j)$, $(i, j-1)$ and $(i, j+1)$ [2].
- J is the **interaction strength** between **neighbouring spins**: $J > 0$ and $J < 0$ corresponding with the tendency for neighbouring spins to align (**ferromagnetism**) or anti-align (**anti-ferromagnetism**), respectively.
- Average magnetisation** of the lattice can be calculated using:

$$\langle m \rangle = \frac{1}{N_x N_y} \sum_{\alpha} S_{\alpha}$$

Scan the QR code to see the lattice evolution of a 32x32 lattice for $B = -0.05$, $J = 0.5$ over 30 sweeps

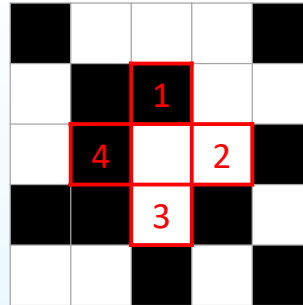


Fig. 1: **Periodic lattice** (i.e. edges wrap around) with randomly assigned spins. White = **spin-up**, Black = **spin-down**. The four **nearest neighbours** of the centre square are shown using red borders and numbers.

2. Partition Function

- Even using a small lattice, checking each possible **state configuration** would be impossible ($2^{N_x N_y}$ configurations)
- To address this, we remove configurations that are extremely unlikely by considering the **partition function**, Z .
- This helps us determine which states are **most probable** at a given temperature, so we can restrict our consideration to these configurations only.
- We define the **partition function** as

$$Z = \sum_i \exp^{E_i/kT}$$

where E_i is the energy of state i , k is the Boltzmann constant, and T is the temperature of the system.

3. Metropolis Algorithm

- To update the lattice, we flip a chosen spin and calculate the resulting **energy change**, $\Delta E = -2[Jf + B]S_{\alpha}$. We use this to gauge how likely this flip is to occur.
 - f is the sum of the **four neighbouring spins**.
 - The negative sign ensures that aligned spins correspond to lower energy.
- We then decide whether to keep the flip using the **Metropolis criterion**:
 - ❖ If $\exp(\Delta E/kT) > 1$ keep the flipped spin
 - ❖ Otherwise, generate a random number, $n \in [0,1]$. If $\exp(-\Delta E/kT) > n$, keep the flip.
 - ❖ If neither condition is met, revert the flip back.
- This procedure is repeated for every lattice point (a **full sweep**), using **periodic boundary conditions**. Together, these steps form the **Metropolis algorithm** [3].

4. Results

- Fig. 2 shows the **average magnetisation** as a function of **sweeps**, for a 32x32 lattice with $B = -0.05$ and $J = 0.5$.
- We define the **steady state region** as the point where the running average magnetisation doesn't fluctuate by more than 5% of the difference between the initial and final value.
- Average magnetisation** predicted by my **simulation**, **Mean Field theory** [4] and an **Ansatz approach** [5] is shown in table 1 below in **Table 1**.

Model	$\langle m \rangle$
Ising model (simulation)	-0.936 ± 0.003
Mean Field Theory	~ -0.96
Ising model (Ansatz)	~ -0.92

Table 1: Average magnetisation predicted by different models

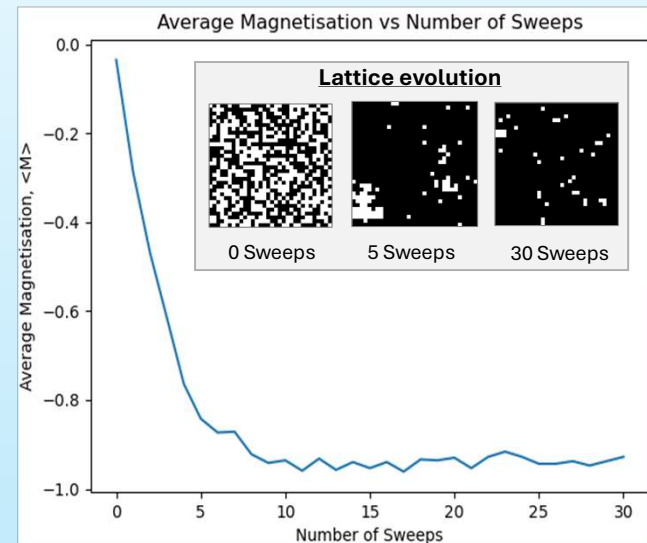


Fig. 2: Plot of average magnetisation per spin against number of sweeps of the Monte Carlo algorithm for a 32x32 lattice, with $B = -0.05$, $J = 0.5$, and $kT = 1$. Insets show lattice at sweeps 0, 5 and 30, further illustrating the rapid evolution into alignment with the B field.

6. References & AI statement

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 - Metropolis, N., Rosenbluth, A., Rosenbluth, M., Teller, A., and Teller, E. (1953). Equations of state calculations by fast computing machines. *J. Chem. Phys.*, 21(6):1087–1092.
 - Sakthivadivel, Dalton A. R. "Magnetisation and mean field theory in the Ising model." *SciPost Physics Lecture Notes*, vol. 35, 2022. arXiv: 2102.00960.
 - Alexanian, Moorad. "Ansatz for the Two-Dimensional Ising Model in an External Magnetic Field." *Armenian Journal of Physics*, vol. 17, no. 3, 2024, pp. 52–57. doi:10.54503/18291171-2024.17.3-52.
- Generative AI in the form of ChatGPT was occasionally used to improve clarity of writing only.



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